AGNIHAWK

Predictive Landfill Fire Prevention System

POWERED BY **PATHWAY** STREAMING ENGINE

Hack For Green Bharat 2026

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THE PROBLEM

India's landfills keep catching fire.

3,159

Landfill fire incidents reported across India (2019–2024)

72hrs

Average burn time at Ghazipur. Toxic fumes for millions in Delhi

500+

PM2.5 µg/m³ during fires. 10× WHO safe limit

Mega-landfills like Ghazipur (Delhi), Deonar (Mumbai), and Brahmapuram (Kochi) sit on top of decades of decomposing waste. Methane builds up. Temperatures rise. One spark — and it burns for days, releasing dioxins, furans, and heavy metals. **There is no early warning system.**

OUR SOLUTION

Predict fires before they start.

Real-Time Sensor Monitoring

Continuous streaming of sub-surface temperature, methane, CO, and weather data across 6 landfill sectors. Every second.

AI Risk Scoring

Streaming risk engine computes a 0–100 fire risk score using weighted sensor fusion. Temperature (40pts), Methane (30pts), CO (15pts), Weather (15pts).

3D Globe Dashboard

CesiumJS interactive globe showing real-time risk across Indian cities. Live AQI, safe evacuation routes, turn-by-turn navigation.

LLM-Powered Alerts

Mistral-7B generates plain-language risk narratives from sensor data. RAG system retrieves fire safety context for accurate recommendations.

ARCHITECTURE

How it works

01

Ingest

3 Pathway streams generate sensor data continuously

`pw.demo.generate_custom_stream()`

→

02

Join

Thermal + Methane joined by sector_id in real-time

`Table.join()`

→

03

Score

Risk engine computes 0-100 score using pw.apply + pw.if_else

`pw.apply()`, `pw.if_else()`

→

04

Push

SSE endpoint streams to browser. Auto-updates.

`pw.io.subscribe()`

→

05

Visualiz

CesiumJS 3D globe shows risk alerts, AQ evacuatio routes

All processing happens in Pathway's streaming engine via `pw.run()` — zero polling, fully reactive

PATHWAY INTEGRATION

14 Pathway APIs, 7 Files

CATEGORY	API	PURPOSE
Connectors	`pw.demo.generate_custom_stream()`	3 sensor streams (thermal, methane, weather)
Schema	`pw.Schema`	4 typed schemas for streaming tables
Transforms	`Table.join()`	Stream join thermal + methane by sector_id
Transforms	`Table.select()`	3 chained selects for feature engineering
Logic	`pw.if_else()`, `pw.apply()`	Risk scoring + level classification
References	`pw.this`	30+ column references in streaming context
Output	`pw.io.subscribe()`	Push updates to Flask SSE via callbacks
Output	`pw.io.jsonlines.write()`	Log alerts to JSONL file
RAG	`pw.xpacks.llm.DocumentStore`	Live fire safety knowledge index
Engine	`pw.run()`	Starts streaming engine — blocks, auto-updates

✅ **Hackathon Rule:** "If your system does not update automatically when new data arrives, it is not a Pathway project." — Our system auto-updates via pw.run(), streaming joins, and pw.io.subscribe(). No polling. No manual refresh.

DASHBOARD FEATURES

What the user sees

3D CesiumJS Globe

Interactive satellite globe with 10 Indian city markers. Color-coded by risk level.

Live Risk Alerts

Real-time alert panel with AI-generated narratives. RED/BLACK alerts trigger evacuation.

3D Buildings

Cesium Ion OSM buildings layer. Photorealistic city skyline visualization.

Live AQI Monitor

City-wide and per-zone air quality. Before/after hazard comparison.

Safe Route Navigation

Google Maps-style turn-by-turn directions. Avoids hazard zones. ETA + distance.

Dark / Light Mode

Tactical dark theme by default. Light mode with OpenStreetMap imagery.

RISK CLASSIFICATION

Five-tier alert system

GREEN	0 – 30	Normal operations. All readings within safe range.
YELLOW	31 – 50	Elevated monitoring. Temperature/methane trending upward.
ORANGE	51 – 70	High risk. Prepare evacuation routes. Deploy pre-wetting teams.
RED	71 – 90	Critical. Begin evacuation. Fire department on standby.
BLACK	91 – 100	Imminent combustion. Full emergency protocol. Evacuate immediately.

Scoring factors: **Temperature** (40pts) + **Methane** (30pts) + **Carbon Monoxide** (15pts) + **Weather Penalty** (15pts) = 100 max

Thank You

Let's prevent the next landfill fire.

GITHUB
github.com/Swagat-CREATOR/AGNIHAWK

LIVE DEMO
localhost:8000

PATHWAY
14 APIs • 7 Files • 100% Streaming

Team AgniHawk — Hack For Green Bharat 2026

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